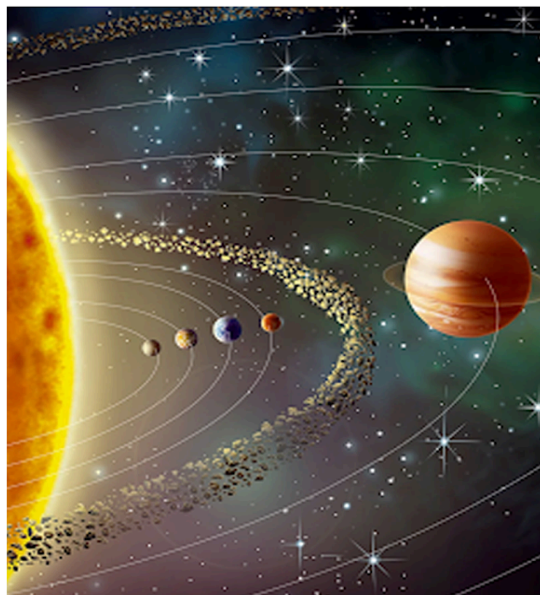
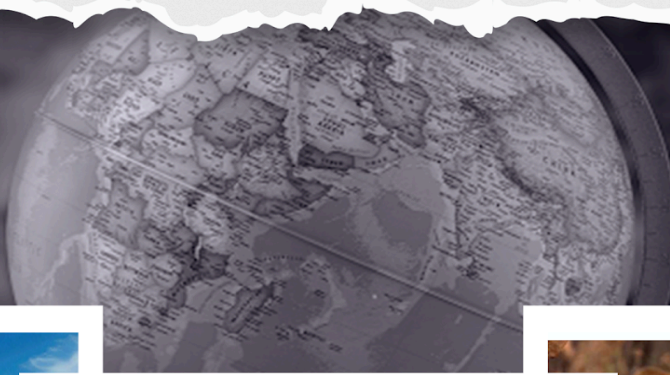


General Knowledge

World Geography

STUDY NOTES

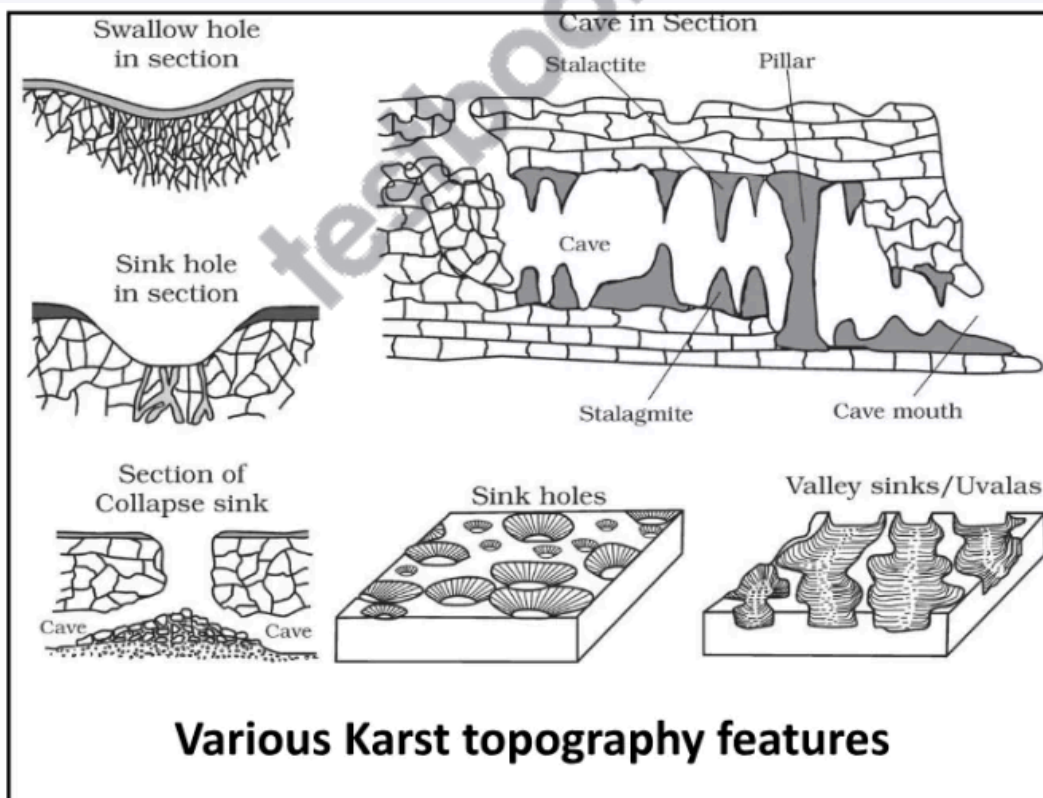
GEOMORPHOLOGY: LANDFORMS(KARST)



GEOMORPHOLOGY: LANDFORMS(KARST)

Karst Landforms and Cycle of Erosion

- Karst landforms are characterized by **underground** drainage systems with **sinkholes, fissures, caves formed from the dissolution (chemical weathering), and erosion of soluble rocks such as limestone, and dolomite.**
- There is a general absence of surface drainage as the water flow is mostly subsurface. In its pure state, limestone is made up of calcium carbonate, but where magnesium is also present, it is termed as **dolomite**.
- **Limestone is an organically formed sedimentary rock** (formed by the decomposition of calcareous shells) and is soluble in rainwater.
- **The carbonic acid that causes karstic features is formed as rain passes through the atmosphere picking up carbon dioxide (CO₂).**
- Once the rain reaches the ground, it may pass through soil that can provide much more CO₂ to form a weak carbonic acid solution, which dissolves calcium carbonate (limestone).
- Karsts are so named after a province of **Yugoslavia** (in the Balkans) **on the Adriatic Sea** coast where such formations are most noticeable.



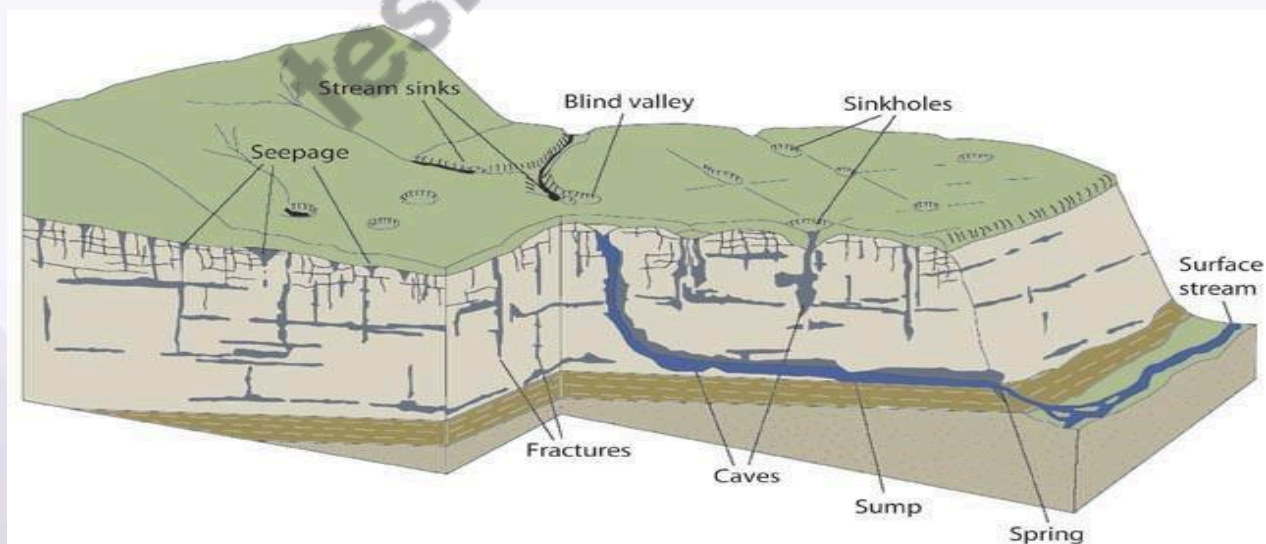
Sinkhole/Swallow Hole

- Sinkholes are funnel-shaped depressions developed by the enlargement of the cracks found in porous water-soluble rocks, as a result of continuous solvent action (chemical weathering) of the rainwater.
- The surface streams disappear underground through swallow holes.
- There is a great variation in the size and depth of sinkholes.



Polje/Blind Valley

- Dolines are small depressions dotting a karst landscape. They are less common than sinkholes
- Some adjoining dolines may come together to form a long, narrow trench called uvala.
- Some uvalas may coalesce to create a 'U' shaped valley called polje.
- If the streams lose themselves in these valleys, then these are called blind valleys.



Cave

- Cave formation is especially prominent in areas where alternating layers of rocks, such as shales, sandstones, and quartzites, are interspersed with limestones or dolomites. Additionally, caves can form in regions where limestones are dense, massive, and occur as thick beds.
- In these areas, water percolates down through the rock layers or along cracks and joints, moving horizontally along the bedding planes.
- It is along these planes that the limestone dissolves, creating narrow or wide gaps that become caves. A network of caves may form at varying elevations, depending on the arrangement of limestone beds and the intervening rock layers.
- These caves often have openings through which cave streams are discharged, and those with openings at both ends are referred to as tunnels.
- In India, such caves can be seen in Bastar, Dehradun, and Shillong plateau.
- **Stalactites** are icicle-like formations that hang from cave ceilings, varying in diameter. They typically have a broad base that tapers towards the free end, taking on various shapes.
- **Stalagmites**, on the other hand, rise from the cave floor, forming from water dripping down from above or through the thin pipe of a stalactite directly above. Stalagmites can have different shapes, such as columns, discs, or rounded bulging tops, sometimes even developing small crater-like depressions.
- Over time, stalagmites and stalactites may merge to form columns and **pillars** of varying sizes.



Sinking Creeks/Bogas

In a karst valley, the water often gets lost through cracks and fissures in the bed. These are called sinking creeks, and if their tops are open, they are called bogas.

